



Victoria Junior College
Biology Department
2017 JC2 Preliminary
Proposed Answers 9744/4

Question 1

(a) (i) Record your observations in a suitable format in the space below, noting particularly any differences that you observe in the appearance of the contents of the five tubes. [6]

- Independent variable must be either contents of tube OR pH in the first column;
- Column heading;
- Observations of contents of tubes – ($\frac{1}{2}$ for extent of fragmentation, $\frac{1}{2}$ for colour / clarity of solution)

Tubes	pH buffer used	Observations of the contents of the tubes
A	pH 2	Egg remained whole / minor fragmentation; Solution remained clear and colourless;
B	pH 4	Egg remained whole / minor fragmentation; Solution remained clear and colourless;
C	pH 6	Egg fragmented slightly (idea of more fragmentation compared to tube A and B); Solution changed from clear colourless to cloudy with a tinge of red;
D	pH 8	Egg fragmented to a greater extent / more extensive fragmentation (compared to tube C); Solution changed from clear colourless to cloudy red;
E	pH X	Egg fragmented slightly (idea of more fragmentation compared to tube A and B); Solution changed from clear colourless to cloudy with a tinge of red;

(ii) Based on your observation in (a) (i), estimate the pH of buffer X. [1]

- pH 6;;

(b) State the conclusions that you can draw, at this stage of the procedure, about the action of K1 on the protein at different pH. Explain how your observations allow you to make these conclusions. [4]

- K1 is an **enzyme** (protease);
- as it is **sensitive to pH changes**;
- K1 enzyme work optimally at **pH8**; (accept ECF based on students' table)
- Idea of **more extensive breakdown of egg** observed in tube D;
- **causing more red dye released** into the solution from egg in tube D;
- When pH goes below its optimum pH, the **conformational shape of enzyme changes**;
- Specific bonds affected - H bonds, ionic bonds, hydrophobic interactions;
- Less ideal for binding of substrate or idea of less ES complexes formed;

- **denaturation** of enzyme;
- Idea of decrease in rate of enzyme reaction;

(c) (i) Discuss two sources of error, other than temperature, that may have affected the accuracy of your results. [2]

- Inconsistency of the shaking action of the tubes; different extent of fragmentation / leakage of stain into the solution may be due to the different strength in shaking;
- Idea of interval of pH too broad as the optimal pH is between the pH interval
- Idea of pH range not extensive enough as the optimal pH may be above pH 8;;
- Staining of egg is not homogenous; affecting the leakage of the stain into the solution;
- Imprecise cutting of egg; leading to different surface area for enzyme to work on;
- Determination of colour and clarity of content is subjective and may deviate amongst individuals;
- Procedures lack the use of a control to show that the effects of breakdown of egg is due to action of K1;

(ii) Suggest improvements to reduce the sources of error you have identified in (c) (i). [2]

- Standardize the method and force for shaking of tubes by using mechanical shaker / vortex;
- Reduce the interval of pH eg. use pH 4, 5, 6, 7, 8
- Extend the range of pH to above 8 eg. use pH 4 to 14
- Use a standardised mold / core borer to cut the egg;
- Use a colorimeter to measure absorbance of light by the contents of the tube;
- Set up a control tube, replacing K1 with same amount of distilled water for all pH and repeat the experiment;

(d) The following experiment in **Fig. 1** was set up to investigate the effect of pH on the activity on the enzyme catalase on potato.

pH buffer	Time taken for potato disc to rise / s			Rate / s ⁻¹
	Disc 1	Disc 2	Average	
4	58	63	60.5	0.0165
5	43	47	45	0.0222
6	41	42	41.5	0.0241
7	39	40	39.5	0.0253
8	44	49	46.5	0.0215
9	56	63	59.5	0.0168

(i) Plot a graph showing the effects of varying pH buffer on the rate of breakdown of hydrogen peroxide in the grid provided below. [4]

- Correct X and Y axis label with units;;
- Scale;;
- Points plotted;;
- Best fit graph plotted;;

(ii) Explain how one can determine that the time taken for potato discs to rise at two different pH is significantly different from each other? [3]

- Repeat the experiment several times (eg. 5-6 times) for each of the two pH;
- Calculate the average time taken for discs to rise at each pH;
- Conduct the T test;
- if $P < 0.05$, then reject the null hypothesis;
- there is significant difference in the timing for discs to rise at the two different pH;
- any difference observed not by chance;
- (accept if students explain $P > 0.05$)

Question 2

(a) You are provided with an extract **P** from plant cells which contains a mixture of different carbohydrates.

You are required to identify which carbohydrates present in **P** can pass through the Visking tubing.

(i) Use the space below to record the tests that you have performed, your observations and conclusions in a suitable format. [4]
Details of the tests are not required.

R= reject; A= Accept

Test for Biomolecules	Observations (Record: change from ___ to ___)	Conclusions
I₂ / KI test for Starch A: iodine test R: starch test	I ₂ /KI turned from brown or yellow to blue black; Accept: dark blue, but not blue	Starch present;
Benedict's test for reducing sugar; (R: B's test for monosaccharides/ glucose as it cannot distinguish between mono and disaccharides. Only detect reducing property)	Contents remained clear blue; (Note: Record both colour and clarity)	Reducing sugar absent; (R: glucose)
Acid hydrolysis followed by Benedict's test (R: Non-reducing sugar test)	Contents turned from clear blue to opaque orange/brick-red; R: cloudy (R: • opaque blue or opaque green – Tr's results is brick red/orange, opaque)	Non-reducing sugars present; (R: sucrose, disaccharide) Note: This test is used to determine the presence of NRS only. It cannot provide identity of the NRS.

(ii) what do your results indicate about the property of the Visking tubing? Explain your answer. [3]

- Property: **semi-permeable/ selectively permeable/ partially permeable;**

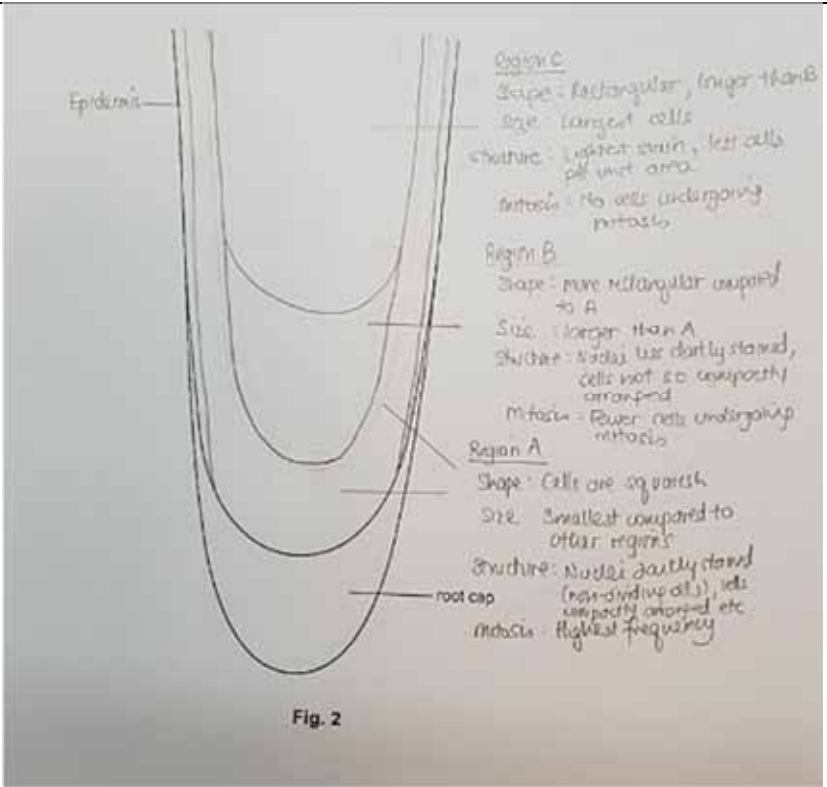
- **Pore size** that only allow small molecules (to diffuse through);
- Non-reducing sugars are able to diffuse through the pores because of their **small** size but starch molecules are too **large** to pass through;;
- Idea of movement (of non-reducing sugars) down a **concentration gradient**;
- Evidence:
 - NRS: blue with tinge of red, cloudy;
 - Starch: remained yellow/brown;

(iii) Explain how you can modify the experiment to determine the **rate of diffusion** of the carbohydrate that you have identified in (a) (ii) above. [3]

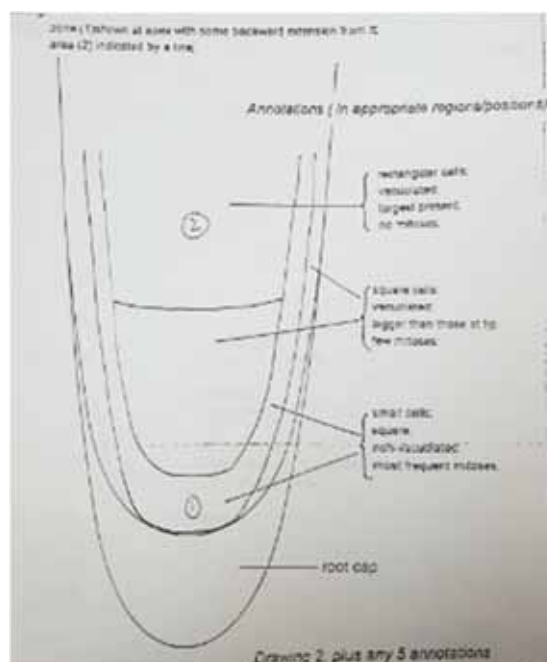
- Sampling (S)
 1. Taking samples at regular time intervals;
 2. 5 time intervals or stated; (A: idea of more than 1 time eg. over 10 mins)
 - Method (M)
 3. **Fix volume** of sample;
 4. Conduct acid hydrolysis and B's test;
 5. Measurement: comparing colour change/ colour and clarity Or measure absorbance/ transmittance using a colorimeter Or determine time taken for first colour change;
 6. determine concentration: by comparing against a colour standard; of known concentration of NRS;
 - Rate (V)
 7. Plot graph of y against (sampling) time;
 8. calculate rate by dividing concentration over time;
 9. Rate taken from gradient;
- {max 3m}

(b) (i) Make a plan drawing of the entire section, within the outline drawn in Fig. 2.1 below to show the different **regions**. These regions result from differences in the **shapes, sizes and structure of the cells** as well as in the **frequency** with which different stages **of mitosis** are visible.

Annotate your drawing as fully as possible to describe the features of the cells in each region that you map. [5]

 Fig. 2 Region C Shape: Rectangular, longer than B Size: Largest cells Structure: Lightest stain, least cells per unit area Mitosis: No cells undergoing mitosis Region B Shape: more rectangular compared to A Size: larger than A Structure: Nuclei less darkly stained, cells not so compactly arranged Mitosis: Fewer cells undergoing mitosis Region A Shape: Cells are squarish Size: Smallest compared to other regions Structure: Nuclei darkly stained (non-dividing cells), cells compactly arranged etc Mitosis: Highest frequency Epidermis root cap	Marking points: 1) Number of layers/zones: at least 3 (not more than 4) 2) Shape and proportion of the 3 layers; 3) Annotation for each layer: 1m x 3 layers (annotation include shape, size and special features eg. dividing cells, position of nucleus etc) 1. Title; 2. Magnification;
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Title: Plan drawing of root tip (DB) seen under 100x magnification
(Note: You are not expected to draw the epidermis)



Another possible plan drawing (J2000)

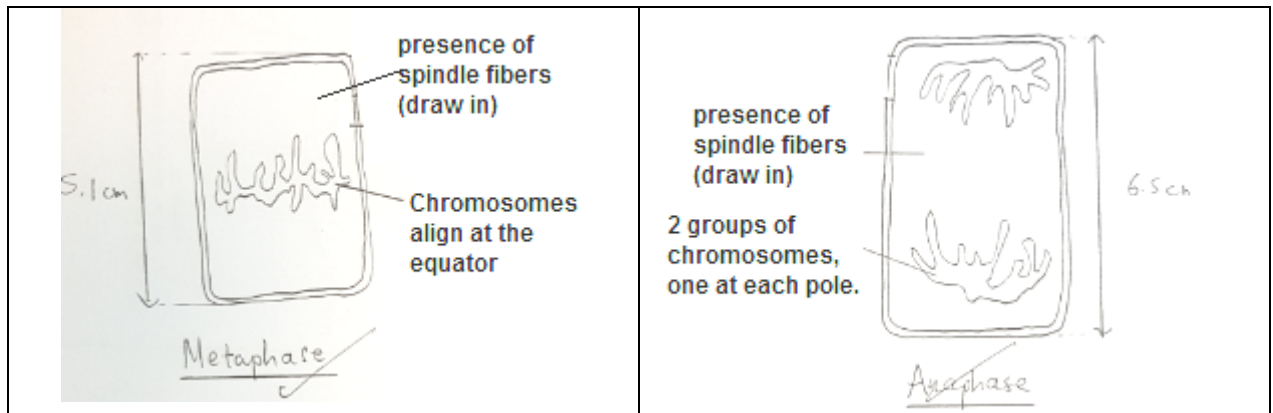
- (ii) In the space below, draw to the **same scale**, two cells that are at different stages of mitosis. Identify the stages and **label the distinctive features** of each stage in your drawings. [5]

Mark scheme

- 1) Same scale: i.e. cells must be relatively the same size and correct shape and proportion;;
- 2) Distinctive features of each stage 1m x2

- 3) Outline : clear, no shading, cell wall drawn as double lines;;
- 4) Magnification;;

Students" work



Question 3

Proposed mark scheme

Theoretical Consideration [2 max]

1. a) Rate of uptake of glucose is higher than rate of uptake of maltose because glucose is the **main respiratory substrate**;
b) Hence **more transport proteins** of glucose present on cell surface membrane OR transport proteins have higher affinity for glucose so increased uptake occurs;
c) Uptake is limited by the number of transport proteins;
2. Outline of strategy and theory supporting it:
a) Method of following the uptake of glucose and maltose **separately**: taking samples at intervals from the suspension and calculating uptake; (A! if students refer to taking one sample after a fixed period of time.)
b) Using Benedict's test to determine the concentration of glucose / maltose / reducing sugars left in the yeast suspension;
3. Hypothesis: Rate of glucose uptake > rate of maltose uptake by their respective transporters at the same concentration.
Predicted results: At every sugar concentration, change in absorbance/time will be higher for glucose (i.e. sharper gradient and higher saturation plateau).

Procedure [7 max]

4. Specify at least 5 different concentrations of glucose and of maltose (e.g. 0, 2, 4, 6, 8, and 10 gdm⁻³) and use of dilution table (glucose / maltose concentrations, vol. of distilled water, and vol. of 10gdm⁻³ glucose / maltose stock solutions must be shown);
5. Factors to be kept constant; @ ½ mark each
a) Temperature (specify a temp. in the range 30 – 40°C)
b) Fixed vol. of 10% yeast suspension
c) Appropriate ratio of sugar solution : yeast suspension (from 5:1 to 10:1 vol./vol. such as 10 cm³ sugar solution to 1 cm³ yeast suspension)
d) Pre-incubation of sugar solutions (e.g. 2-5min) to attain temp. of water bath;
6. Logical sequence of procedure steps:
a) Yeast mixed with sugar solution;
b) Sampling of glucose or maltose at fixed intervals (specify at least 5 regular time intervals e.g. 0, 5, 10, 15, 20 minutes);
c) Filtered / centrifuged to remove yeast;
6. Description of Benedict's test for the presence of glucose / maltose (e.g. boiling water bath for 2-5 minutes); colour and clarity;
7. Description of use of colorimeter; for determining the concentration of glucose / maltose based on absorbance value (note absorbance units is in A);
8. Showing calculation of quantity of sugars taken in, based on difference between initial and final concentration.
9. Description of a control: same conditions (e.g. use same vol. / vol. ratio using 10gdm⁻³ glucose / maltose (how) but using boiled and cooled yeast (what) to show that functional transport proteins are required for the uptake of sugar (why));
10. Repeats and replicates (repeat the whole experiment twice with replicates incorporated in the write up);;

Results [2]

11. Table: use of correct headings and processed to average absorbance value of glucose / maltose and rate of uptake of glucose / maltose;;
12. Plot graph (Y axis: rate of uptake of sugar / A min^{-1} , X axis: concentration of sugar / g dm^{-3}) that show the kinetics of both glucose and maltose;;

Risk consideration [1]

What, why, how: Benedict's solution irritates the skin, wearing of gloves and goggles to avoid contact; AVP: yeast solution, HCl if it is used